

Lab Assignment: Inheritance, Encapsulation, and Collections in a Two-Character Duel

Scenario and Rules

Two characters — a **Wizard** and a **Warrior** — duel by alternately attacking each other.

1. **Damage rule (defender only):** After each attack, the attacked character's **health** is reduced by a random integer in the range $[0, 10]$.
2. **Healing trigger:** If the damage is 0, 5, or 10, the attacked character immediately heals by a random integer in $[0, 10]$.
3. **Attacker's resource reduction (random, relative to initial resource):**
 - After every attack, the attacker reduces its own resource by a random amount.
 - **Wizard:** reduce mana by $rand \in [0, initialMana]$.
 - **Warrior:** reduce strength by $rand \in [0, initialStrength]$.
 - Resources cannot become negative.
4. **Duel rule (termination):** Two characters duel by attacking each other; the first whose *health*, *strength*, or *mana* reaches 0 loses.
 - Wizard loses if `health == 0` or `mana == 0`.
 - Warrior loses if `health == 0` or `strength == 0`.
 - The duel stops immediately when the first losing condition is met.

Class Specifications

1) GameCharacter (Superclass)

- **Private fields:** `String name`, `int health`.
- **Constructor:** validates `health` (≥ 0).
- **Methods:**
 - `getName()`, `getHealth()`, `setHealth(int h)`.
 - `attack(GameCharacter target)`: applies random damage and conditional healing.
 - `takeDamage(int)`, `heal(int)`, `toString()`.

2) Wizard (Subclass of GameCharacter)

- **Private fields:** `int mana`, `final int initialMana`.
- **Constructor:** calls `super`, sets `mana`, and stores `initial mana`.
- **Methods:**
 - `getMana()`, `setMana(int)` with range $[0, 100]$.
 - Override `attack`: call `super.attack()`, then reduce `mana` by random $[0, initialMana]$.

3) Warrior (Subclass of GameCharacter)

- **Private fields:** `int strength`, `final int initialStrength`.
- **Constructor:** calls `super`, sets `strength`, and stores initial strength.
- **Methods:**
 - `getStrength()`, `setStrength(int)` ensuring ≥ 0 .
 - Override `attack`: call `super.attack()`, then reduce strength by random $[0, initialStrength]$.

Program Requirements (Main Program)

1. Prompt the user for Wizard and Warrior details (name, health, mana/strength).
2. Store both characters in an `ArrayList<GameCharacter>`.
3. Print initial status.
4. Run duel loop:
 - Alternate turns (Wizard $\rightarrow$ Warrior $\rightarrow$ Wizard $\rightarrow$...).
 - After each attack, print attack message and updated status.
 - Check losing conditions and stop immediately when reached.
5. Print final result.

Submission Instructions

- **Platform:** Google Classroom
- **Submit:**
 - Source files: `GameCharacter.java`, `Wizard.java`, `Warrior.java`, `DuelRPG.java`.
 - Three screenshots showing **different duel outcomes**, each until the end of the game (final winner announcement included).

Example Console Output (Complete Run)

```
Enter Wizard name: Merlin
Enter Wizard health: 40
Enter Wizard mana (0-100): 50
Enter Warrior name: Thor
Enter Warrior health: 55
Enter Warrior strength (>=0): 12

--- Initial Status ---
Name: Merlin, Health: 40, Mana: 50
Name: Thor, Health: 55, Strength: 12
```

--- Duel Begins ---

Merlin attacks Thor for 7 damage. [Wizard mana -12, now 38]

Status -> Name: Merlin, Health: 40, Mana: 38

Status -> Name: Thor, Health: 48, Strength: 12

Thor attacks Merlin for 10 damage. (Triggered heal: Merlin heals 3)

[Warrior strength -4, now 8]

Status -> Name: Merlin, Health: 33, Mana: 38

Status -> Name: Thor, Health: 48, Strength: 8

Merlin attacks Thor for 0 damage. (Triggered heal: Thor heals 6)

[Wizard mana -7, now 31]

Status -> Name: Merlin, Health: 40, Mana: 31

Status -> Name: Thor, Health: 54, Strength: 8

Thor attacks Merlin for 5 damage. (Triggered heal: Merlin heals 2)

[Warrior strength -8, now 0]

Status -> Name: Merlin, Health: 37, Mana: 31

Status -> Name: Thor, Health: 54, Strength: 0

--- Duel Ends ---

Thor loses (health or strength reached 0). Merlin wins.